



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

KAFKA STREAMS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

Q1 What is Kafka Streams and what problem in Data Engineering does it solve?

Q6 How does Kafka Streams handle reliability and high availability?

Q2 What are the core components or concepts in Kafka Streams?

Q7 What observability does Kafka Streams provide out of the box?

Q3 How does Kafka Streams compare to its main alternatives?

Q8 What are common performance bottlenecks in Kafka Streams?

Q4 How do you get started with Kafka Streams for a new project?

Q9 What security best practices apply when running Kafka Streams?

Q5 What configuration options most affect Kafka Streams's behavior in production?

Q10 What mistakes do teams commonly make when adopting Kafka Streams?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Kafka Streams is a tool or platform

Mitigation

Kafka Streams is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

A6 Kafka Streams provides HA through leader election, Observability

Kafka Streams provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Kafka Streams has key building blocks that

Primitives

Kafka Streams has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Kafka Streams exposes metrics (Prometheus endpoint or information

Kafka Streams exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Kafka Streams trades off simplicity against feature

Hardening

Kafka Streams trades off simplicity against feature richness differently than its alternatives. Choose Kafka Streams when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Kafka Streams via its package manager

Gotchas

Install Kafka Streams via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Kafka Streams with the principle of

Assessment

Run Kafka Streams with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency, configuration

Concurrency

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and organization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.